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Ryan Thompson edited this page 2 weeks ago · 83 revisions

Welcome to the bladeRF wiki!

This wiki will serve as a place to create some community-based documentation. Please feel free to create a page and link it here!

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Getting Started

bladeRF software build/installation

These guides describe the process of building and installing the bladeRF software from source code. If you've just received your bladeRF, this is the place to start.

- Getting Started: Linux Live Images
- Getting Started: Linux
- Getting Started: Windows
- Getting Started: Mac OSX
- Getting Started: Verifying Basic Device Operation

bladeRF firmware images

- FX3 firmware images
- FPGA images

Expansion Boards and Accessories

- Getting Started: XB-200 Transverter Board
- Getting Started: XB-300 Amplifier Board
- Getting Started: BT-100 Bias-Tee Power Amplifier
- Getting Started: BT-200 Bias-Tee Low-Noise Amplifier
- bladeRF Accessories

Developing with libbladeRF

- libbladeRF API documentation

Help and How-to's

- Troubleshooting
  - This guide lists common problems and issues, with some advice and potential fixes.
- Upgrading the bladeRF firmware
  - This guide describes both the normal firmware and bootloader-based firmware upgrade procedures.
- bladeRF CLI Tips and Tricks
  - This pages provides information regarding the use of the bladeRF-cli
- FPGA Autoloading
  - Setting up the bladeRF to have its FPGA loaded automatically
- DC offset and IQ Imbalance Correction
  - Information about DC offset and IQ balance correction mechanisms
- Debugging dropped samples and identifying achievable sample rates
- Soapy SDR plugin for the bladeRF

    - See the SoapySDR project for additional information. Be sure to check out the Python bindings!
- FX3 Firmware
    - Information about FX3 firmware development
- FPGA Development
    - Information about Altera Cyclone IV FPGA development

Projects, Papers, and Blogs

Feel free to share your favorite projects, papers, blogs, etc. containing bladeRF and/or SDR content! Be sure to also check out the Project Showcase on the forums.

- Null Team's Yate & YateBTS
    - Setting up Yate and YateBTS with the bladeRF
    - Yate Blog: Software Defined Mobile Network
        - OFDM - the science behind LTE
- OpenAirInterface
- Software Radio System
    - srsUE -- bladeRF support added in 1.1
    - srsLTE
- OpenBTS
    - Minimalistic build and run test for OpenBTS 5
    - "Should you need OpenBTS on your bladeRF" on the *The Sorrows of Interactive Media* blog
- ATSC Transmitter
- Blog: Clayton's Domain
    - Reverse Engineering a Ceiling Fan
    - Digital amateur TV on 70cm, 33cm, and 23cm
- GitHub: argilo's sdr-examples
- GitHub: drmpeg's projects
    - Utilities for SDR digital television
    - A DVB-S transmitter for GNU Radio
    - A DVB-S2 and DVB-S2X transmitter for GNU Radio
    - Mapping block for bladeRF ATSC modulater and RRC filter FPGA image
- GitHub: GPS baseband signal generator
- GitHub: GNSS-SDRLIB: An Open Source GNSS Software Defined Radio Library
- DEF CON 23 Slides: GPS Spoofing: Low Cost GPS Simulator - Huang Lin & Yang Qing
- Jiao Xianjun (BH1RXH)'s tech blog and publications
    - Fork of an OpenCL accelerated TDD/FDD LTE Scanner, with bladeRF support added

    - A BTLE (Bluetooth Low energy)/BT4.0 radio packet transmitter
- GitHub: rtl-entropy: An entropy generator using SDR peripherals, including rtl-sdr and BladeRF
- Blog Post by Sean Cassidy: Your Own Verifiable Hardware RNG with bladeRF SDR
- An inter-balloon data relay using Software Defined Radios: Paving the way for distributed space systems
    - Congratulations to team MONSTER for winning the "Best Space Technology Demonstration" prize in the Global Space Balloon Challenge!
- Implementation of a Low-Latency Contention-Free Geographical Routing Scheme for Mobile Cyber-Physical Systems
- Microsoft Research:
    - Ziria: Wireless Programming for Hardware Dummies
    - Demo of an 802.11a/g sniffer implemented in Ziria
- Beep Networks' Spectrum Viewer
    - Blog Post: Fun with Software-Defined Radios
- Pothosware
    - SoapySDR - Vendor and platform neutral SDR support library
    - QPSK Modem Demo
    - SoapySDR support for bladeRF timestamps
- RAMEAR - Receive signals from an air-gapped computer
    - PennApps 2016 winner based upon GSMem paper
    - GitHub project page
- Software-Defined Radar for Medical Imaging

Tutorials and Related Reading

Find or learn something interesting? Share it here!

Signal Processing and RF

- Quadrature Signals: Complex, But Not Complicated
- I/Q Data for Dummies
- MIT OpenCourseWare: Signals and Systems (Spring 2011), Prof Alan V. Oppenheim
- Complete GPS/GLONASS Receiver Design
- Software Radio for Experimenters with GNU Radio, Octave and Python by Michel Barbeau
- Let's Assume the System Is Synchronized by Fredric J. Harris
- Presentations:
    - Multirate Digital Signal Processing by Fredric J. Harris

- Videos:
  - Basics of IQ Signals Part I
  - Basics of IQ Signals Part II
- Books:
  - Understanding Digital Signal Processing by Richard G. Lyons
  - Software Receiver Design: Build Your Own Digital Communication System In Five Easy Steps by C. Richard Johnson, Jr., William A. Sethares, & Andrew G. Klein
- Application Notes:
  - Atmel: Manchester Coding Basics

Amateur Radio

- American Radio Relay League
- US Amateur Radio Bands

Development

See the doc/ and doc/development directories for "official" development-related documentation.

Recommended Reading

- Using Git and GitHub
- Debugging

Project Maintenance

- Testing
- Linux Distribution Status
- Static Analysis
- Proposals and Requests
- FX3 Firmware Development

Ongoing and future efforts

Working on new functionality or adding bladeRF support to a project? Create a page here to plan and document your efforts.

- Current and planned development tasks
- OpenBSC

Clone this wiki locally

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https://github.com/Nuand/bladeRF.wiki.git
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